



Cloud Computing

BSE-VB

Submitted By

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Submitted to

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LAB-05

Lab 5 – Java, apt vs apt-get, snap, GUI, Vim on Ubuntu Server

Objective

This lab focuses on package discovery and management on Ubuntu Server, installing and configuring a lightweight GUI for remote access, and practicing Vim editing and navigation.

Task 1: Discover missing command & install Java using apt suggestion

Explanation:

You will run the java command to see what package Ubuntu suggests for installation, then install, verify, and remove it.

Steps with Required Screenshots:

1. Run java to see package suggestion.
Screenshot: task1_java_suggestion.png

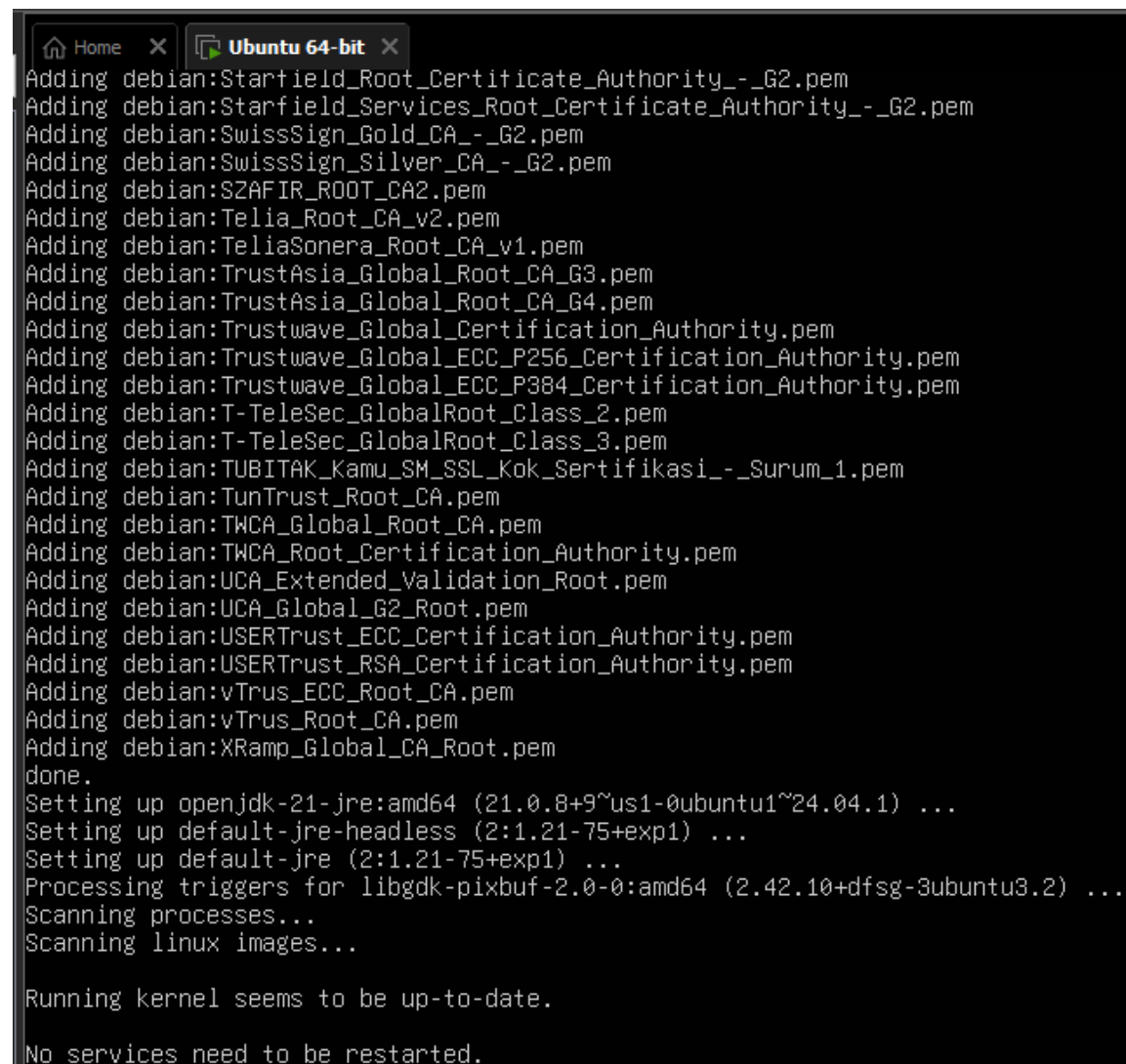
```

validity:it forever preferred:it forever
tooba65@ubuntu:~$ java
Command 'java' not found, but can be installed with:
sudo apt install openjdk-17-jre-headless # version 17.0.16+8~us1-0ubuntu1~24.04.1, or
sudo apt install openjdk-21-jre-headless # version 21.0.8+9~us1-0ubuntu1~24.04.1
sudo apt install default-jre # version 2:1.17-75
sudo apt install openjdk-11-jre-headless # version 11.0.28+6-1ubuntu1~24.04.1
sudo apt install openjdk-25-jre-headless # version 25+36-1~24.04.2
sudo apt install openjdk-8-jre-headless # version 8u462-ga~us1-0ubuntu2~24.04.2
sudo apt install openjdk-19-jre-headless # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless # version 22~22ea-1

```

2. Install suggested package using apt.

Screenshot: task1_java_install.png



```

Home x Ubuntu 64-bit x
Adding debian:Startfield_Root_Certificate_Authority_-_G2.pem
Adding debian:Starfield_Services_Root_Certificate_Authority_-_G2.pem
Adding debian:SwissSign_Gold_CA_-_G2.pem
Adding debian:SwissSign_Silver_CA_-_G2.pem
Adding debian:SZAFIR_ROOT_CA2.pem
Adding debian:Telia_Root_CA_v2.pem
Adding debian:TeliaSonera_Root_CA_v1.pem
Adding debian:TrustAsia_Global_Root_CA_G3.pem
Adding debian:TrustAsia_Global_Root_CA_G4.pem
Adding debian:Trustwave_Global_Certification_Authority.pem
Adding debian:Trustwave_Global_ECC_P256_Certification_Authority.pem
Adding debian:Trustwave_Global_ECC_P384_Certification_Authority.pem
Adding debian:T-TeleSec_GlobalRoot_Class_2.pem
Adding debian:T-TeleSec_GlobalRoot_Class_3.pem
Adding debian:TUBITAK_Kamu_SM_SSL_Kok_Sertifikasi_-_Surum_1.pem
Adding debian:TunTrust_Root_CA.pem
Adding debian:TWCA_Global_Root_CA.pem
Adding debian:TWCA_Root_Certification_Authority.pem
Adding debian:UCA_Extended_Validation_Root.pem
Adding debian:UCA_Global_G2_Root.pem
Adding debian:USERTrust_ECC_Certification_Authority.pem
Adding debian:USERTrust_RSA_Certification_Authority.pem
Adding debian:vTrus_ECC_Root_CA.pem
Adding debian:vTrus_Root_CA.pem
Adding debian:XRamp_Global_CA_Root.pem
done.
Setting up openjdk-21-jre:amd64 (21.0.8+9~us1-0ubuntu1~24.04.1) ...
Setting up default-jre-headless (2:1.21-75+exp1) ...
Setting up default-jre (2:1.21-75+exp1) ...
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.42.10+dfsg-3ubuntu3.2) ...
Scanning processes...
Scanning linux images...

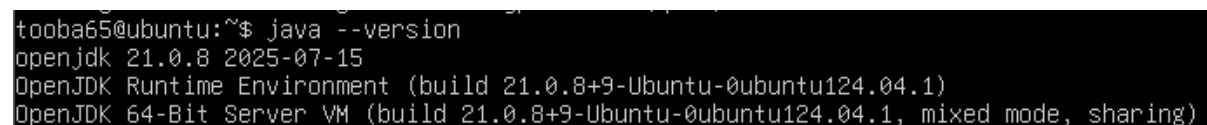
Running kernel seems to be up-to-date.

No services need to be restarted.

```

3. Verify installation using java --version.

Screenshot: task1_java_version.png



```

tooba65@ubuntu:~$ java --version
openjdk 21.0.8 2025-07-15
OpenJDK Runtime Environment (build 21.0.8+9-Ubuntu-0ubuntu124.04.1)
OpenJDK 64-Bit Server VM (build 21.0.8+9-Ubuntu-0ubuntu124.04.1, mixed mode, sharing)

```

4. Remove package using apt remove.

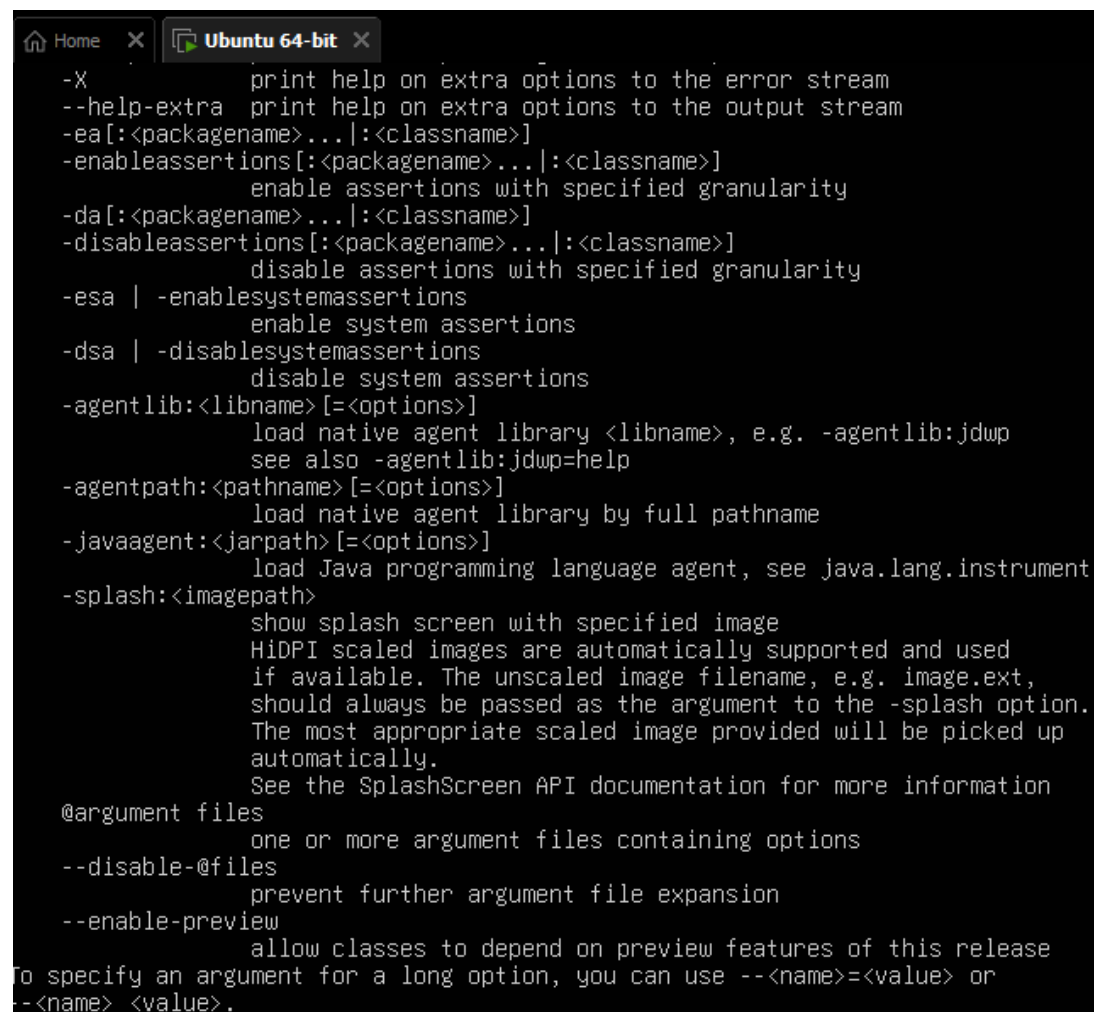
Screenshot: task1_java_remove.png

```
tooba65@ubuntu:~$ sudo apt remove default-jre -y

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificat
  default-jre-headless fontconfig fonts-dejavu-extra gsettings-desktop-schemas gtk-update-icon-
  libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-j
  libavahi-common-data libavahi-common3 libcairo-gobject2 libcairo2 libcolord2 libcups2t64 libd
  libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgl1 libgl1-mesa-d
  libgtk-3-bin libgtk-3-common libharfbuzz0b libice6 liblcms2-2 libllvm20 libpango-1.0-0 libpan
  libpixman-1-0 librsvg2-2 librsvg2-common libsm6 libthai-data libthai0 libvulkan1 libwayland-c
  libx11-xcb1 libxaw7 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-render0 li
  libxcomposite1 libxcursor1 libxdamage1 libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1 lib
  libxv1 libxxf86dga1 libxxf86vm1 mesa-libgallium mesa-vulkan-drivers openjdk-21-jre openjdk-21
  x11-utils
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  default-jre
0 upgraded, 0 newly installed, 1 to remove and 28 not upgraded.
After this operation, 6,144 B disk space will be freed.
(Reading database ... 101619 files and directories currently installed.)
Removing default-jre (2:1.21-75+exp1) ...
```

5. Confirm Java is unavailable again.

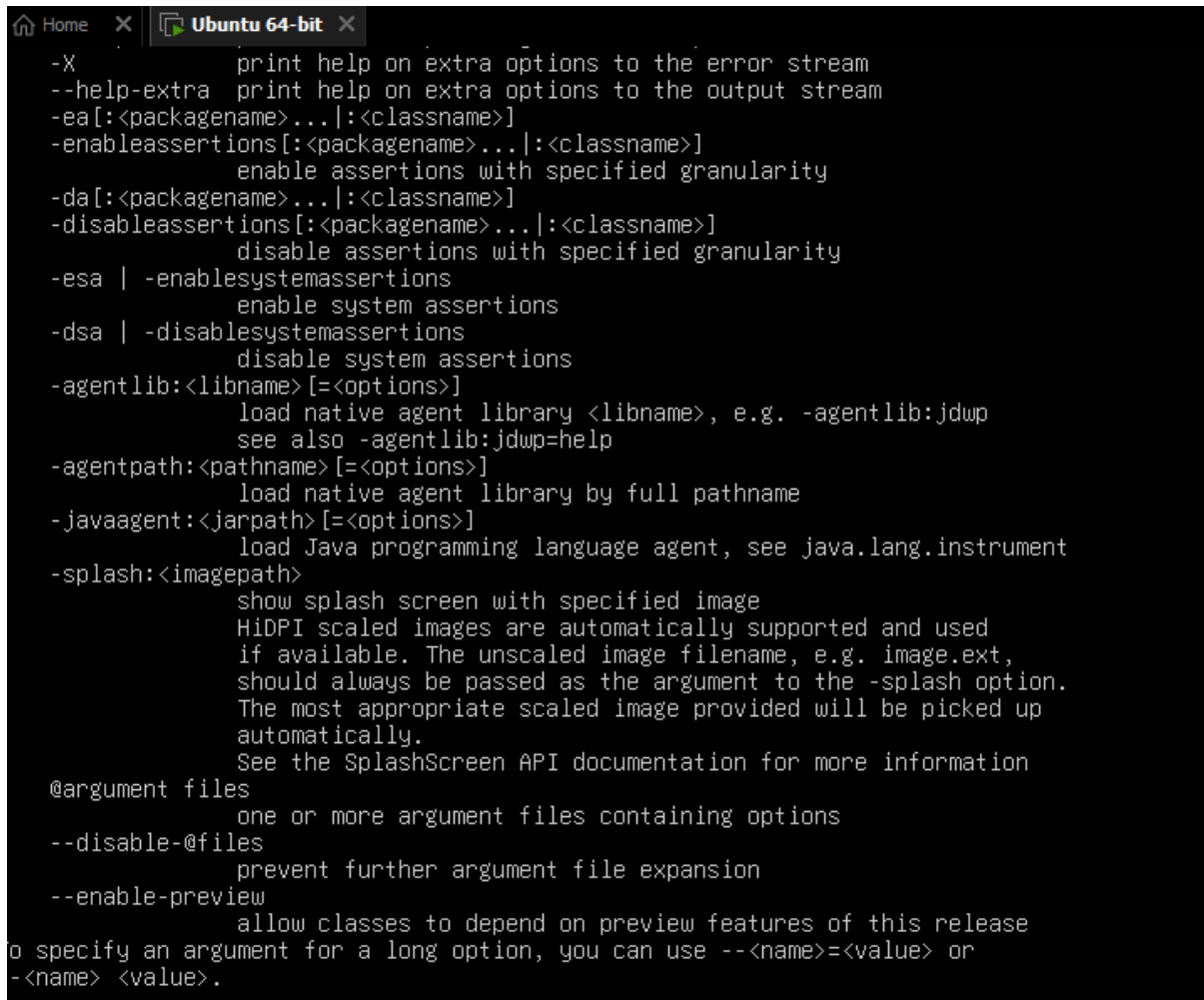
Screenshot: task1_java_not_found.png



```
Home x Ubuntu 64-bit x
-X          print help on extra options to the error stream
--help-extra print help on extra options to the output stream
-ea[:<packagename>...[:<classname>]]
-enableassertions[:<packagename>...[:<classname>]]
            enable assertions with specified granularity
-da[:<packagename>...[:<classname>]]
-disableassertions[:<packagename>...[:<classname>]]
            disable assertions with specified granularity
-esa | -enablesystemassertions
            enable system assertions
-dsa | -disablesystemassertions
            disable system assertions
-agentlib:<libname>[=<options>]
            load native agent library <libname>, e.g. -agentlib:jdwp
            see also -agentlib:jdwp=help
-agentpath:<pathname>[=<options>]
            load native agent library by full pathname
-javaagent:<jarpath>[=<options>]
            load Java programming language agent, see java.lang.instrument
-splash:<imagepath>
            show splash screen with specified image
            HiDPI scaled images are automatically supported and used
            if available. The unscaled image filename, e.g. image.ext,
            should always be passed as the argument to the -splash option.
            The most appropriate scaled image provided will be picked up
            automatically.
            See the SplashScreen API documentation for more information
@argument files
            one or more argument files containing options
--disable-@files
            prevent further argument file expansion
--enable-preview
            allow classes to depend on preview features of this release
To specify an argument for a long option, you can use --<name>=<value> or
--<name> <value>.
```

6. Clear shell cache using hash -r and test again.

Screenshot: task1_hash_clear.png



```
-X          print help on extra options to the error stream
--help-extra print help on extra options to the output stream
-ea[:<packagename>...[:<classname>]]
-enableassertions[:<packagename>...[:<classname>]]
            enable assertions with specified granularity
-da[:<packagename>...[:<classname>]]
-disableassertions[:<packagename>...[:<classname>]]
            disable assertions with specified granularity
-esa | -enablessystemassertions
            enable system assertions
-dsa | -disablesystemassertions
            disable system assertions
-agentlib:<libname>[=<options>]
            load native agent library <libname>, e.g. -agentlib:jdwp
            see also -agentlib:jdwp=help
-agentpath:<pathname>[=<options>]
            load native agent library by full pathname
-javaagent:<jarpath>[=<options>]
            load Java programming language agent, see java.lang.instrument
-splash:<imagepath>
            show splash screen with specified image
            HiDPI scaled images are automatically supported and used
            if available. The unscaled image filename, e.g. image.ext,
            should always be passed as the argument to the -splash option.
            The most appropriate scaled image provided will be picked up
            automatically.
            See the SplashScreen API documentation for more information
@argument files
            one or more argument files containing options
--disable-@files
            prevent further argument file expansion
--enable-preview
            allow classes to depend on preview features of this release
o specify an argument for a long option, you can use --<name>=<value> or
-<name> <value>.
```

Task 2: Install & remove Java using apt-get

Explanation:

Repeat Java installation and removal using apt-get to understand the difference between the two package managers.

Steps with Required Screenshots:

1. Install Java using apt-get.

Screenshot: task2_aptget_install.png

```

Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,260 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,536 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [292 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.3 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,164 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [208 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [8,984 B]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,069 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [488 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,496 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [301 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.3 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:22 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,148 B]
Get:23 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:24 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:26 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [468 kB]
Get:27 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:28 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [500 B]
Get:29 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [904 kB]
Get:30 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [202 kB]
Get:31 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.3 kB]
Get:32 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.3 kB]
Get:33 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [27.4 kB]
Get:34 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [5,708 B]
Get:35 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:36 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [384 B]
Fetched 12.5 MB in 19s (649 kB/s)
Reading package lists... Done

```

```

Get:32 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.3 kB]
Get:33 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [27.4 kB]
Get:34 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [5,708 B]
Get:35 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:36 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [384 B]
Fetched 12.5 MB in 19s (649 kB/s)
Reading package lists... Done
tooba65@ubuntu:~$ sudo apt-get install default-jre -y

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  default-jre
0 upgraded, 1 newly installed, 0 to remove and 35 not upgraded.
Need to get 922 B of archives.
After this operation, 6,144 B of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 default-jre amd64 2:1.21-75+exp1 [922 B]
Fetched 922 B in 0s (6,401 B/s)
Selecting previously unselected package default-jre.
(Reading database ... 101618 files and directories currently installed.)
Preparing to unpack .../default-jre_2%3a1.21-75+exp1_amd64.deb ...
Unpacking default-jre (2:1.21-75+exp1) ...
Setting up default-jre (2:1.21-75+exp1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```

2. Verify version using `java --version`.

Screenshot: task2_java_version_after_aptget.png

```
tooba65@ubuntu:~$ java --version
openjdk 21.0.8 2025-07-15
OpenJDK Runtime Environment (build 21.0.8+9-Ubuntu-0ubuntu124.04.1)
OpenJDK 64-Bit Server VM (build 21.0.8+9-Ubuntu-0ubuntu124.04.1, mixed mode, sharing)
```

3. Remove package using `apt-get remove`.

Screenshot: task2_aptget_remove.png

```
tooba65@ubuntu:~$ sudo apt-get remove default-jre -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dconf-
  default-jre-headless fontconfig fonts-dejavu-extra gsettings-desktop-schemas gtk-update-icon-cache hicolor-
  libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0
  libavahi-common-data libavahi-common3 libcairo-gobject2 libcairo2 libcolorl2 libcupss2t64 libdatrie1 libdconf
  libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgl1 libgl1-mesa-dri libglvnd0 l
  libgtk-3-bin libgtk-3-common libharfbuzz0b libice6 liblcms2-2 libllvm20 libpango-1.0-0 libpangocairo-1.0-0
  libpixman-1-0 librsync2-2 librsync2-common libsm6 libthai-data libthai0 libvulkan1 libwayland-client0 libwayl
  libx11-xcb1 libxaw7 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-render0 libxcb-shape0 li
  libxcomposite1 libxcursor1 libxdamage1 libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1 libxmu6 libxrandr
  libxv1 libxxf86dga1 libxxf86vm1 mesa-libgallium mesa-vulkan-drivers openjdk-21-jre openjdk-21-jre-headless
  x11-utils
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  default-jre
0 upgraded, 0 newly installed, 1 to remove and 35 not upgraded.
After this operation, 6,144 B disk space will be freed.
(Reading database ... 101619 files and directories currently installed.)
Removing default-jre (2:1.21-75+exp1) ...
```

4. Clear command cache and verify.

Screenshot: task2_hash_after_remove.png

```
Home x Ubuntu 64-bit x
-X                print help on extra options to the error stream
--help-extra      print help on extra options to the output stream
-ea[:<packagename>...[:<classname>]]
  enable assertions with specified granularity
-da[:<packagename>...[:<classname>]]
  disable assertions with specified granularity
-esa | -enablesystemassertions
  enable system assertions
-dsa | -disablesystemassertions
  disable system assertions
-agentlib:<libname>[=<options>]
  load native agent library <libname>, e.g. -agentlib:jdwp
  see also -agentlib:jdwp=help
-agentpath:<pathname>[=<options>]
  load native agent library by full pathname
-javaagent:<jarpath>[=<options>]
  load Java programming language agent, see java.lang.instrument
-splash:<imagepath>
  show splash screen with specified image
  HiDPI scaled images are automatically supported and used
  if available. The unscaled image filename, e.g. image.ext,
  should always be passed as the argument to the -splash option.
  The most appropriate scaled image provided will be picked up
  automatically.
  See the SplashScreen API documentation for more information
@argument files
  one or more argument files containing options
--disable-@files
  prevent further argument file expansion
--enable-preview
  allow classes to depend on preview features of this release
To specify an argument for a long option, you can use --<name>=<value> or
--<name> <value>.
```

Task 3: apt update vs apt upgrade

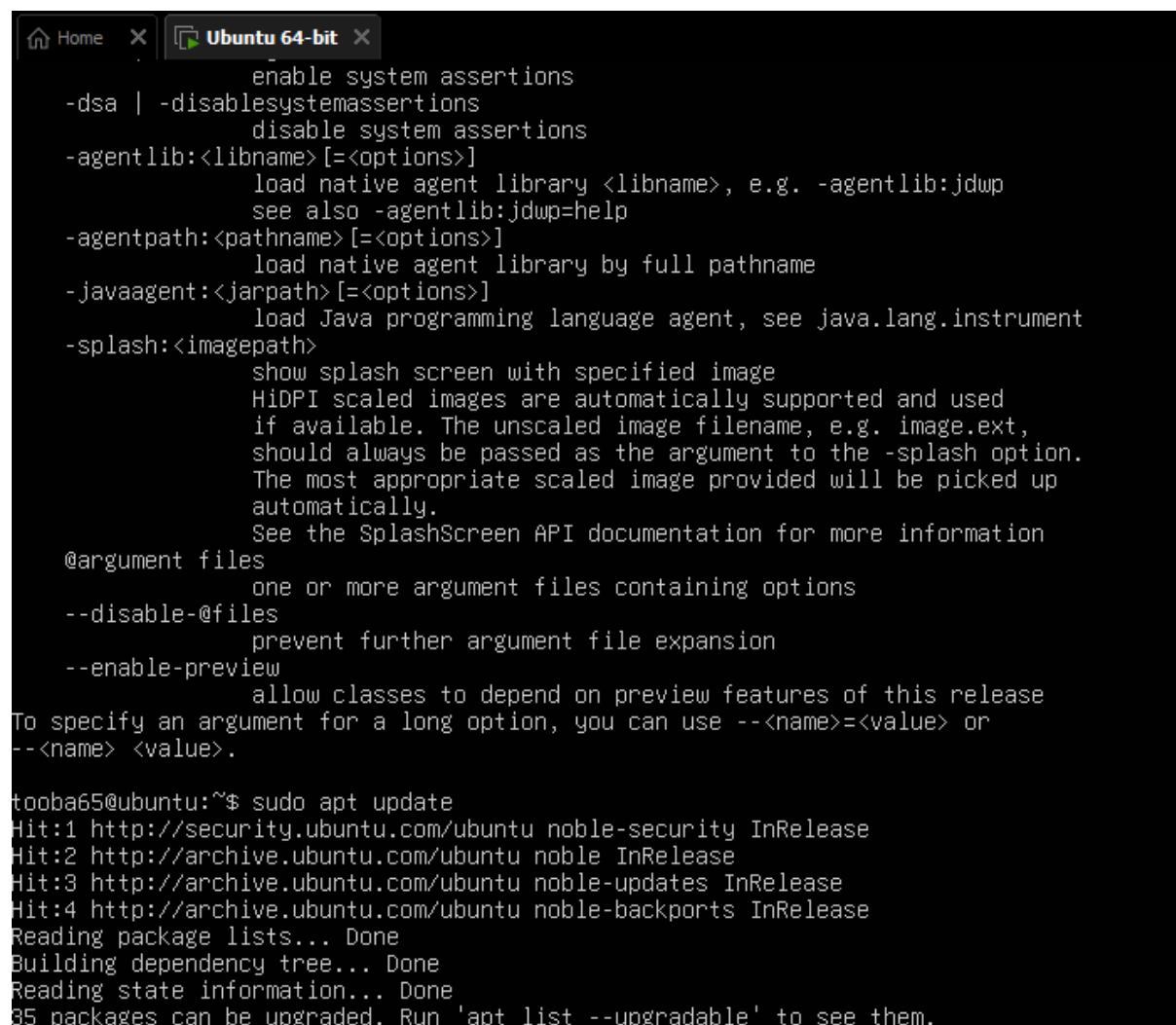
Explanation:

You will learn the difference between updating package indexes and upgrading installed packages.

Steps with Required Screenshots:

1. Run apt update.

Screenshot: task3_apt_update.png

A terminal window titled 'Ubuntu 64-bit' showing the execution of the 'apt update' command. The terminal displays the help text for the 'apt' command, followed by the command 'sudo apt update' and its output. The output shows four hits for security and archive updates, followed by 'Reading package lists... Done', 'Building dependency tree... Done', 'Reading state information... Done', and a final message stating '35 packages can be upgraded. Run 'apt list --upgradable' to see them.'

```
enable system assertions
-dsa | -disablesystemassertions
        disable system assertions
-agentlib:<libname>[=<options>]
        load native agent library <libname>, e.g. -agentlib:jdwp
        see also -agentlib:jdwp=help
-agentpath:<pathname>[=<options>]
        load native agent library by full pathname
-javaagent:<jarpath>[=<options>]
        load Java programming language agent, see java.lang.instrument
-splash:<imagepath>
        show splash screen with specified image
        HiDPI scaled images are automatically supported and used
        if available. The unscaled image filename, e.g. image.ext,
        should always be passed as the argument to the -splash option.
        The most appropriate scaled image provided will be picked up
        automatically.
        See the SplashScreen API documentation for more information
@argument files
        one or more argument files containing options
--disable-@files
        prevent further argument file expansion
--enable-preview
        allow classes to depend on preview features of this release
To specify an argument for a long option, you can use --<name>=<value> or
--<name> <value>.

tooba65@ubuntu:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
35 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

2. Run apt upgrade.

Screenshot: task3_apt_upgrade.png

```
Home x Ubuntu 64-bit x
Found linux image: /boot/vmlinuz-6.8.0-85-generic
Found initrd image: /boot/initrd.img-6.8.0-85-generic
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-85-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-86-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...
systemctl restart multipathd.service open-vm-tools.service polkit.service ssh.service udisks2.service upower.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
tooba65 @ session #1: apt[3903], login[920]
tooba65 @ session #3: sshd[1276]
tooba65 @ user manager service: systemd[1204]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
tooba65@ubuntu:~$
```

3. Write and save explanation in a text file.

Screenshot: task3_explanation.png

```
tooba65@ubuntu:~$ cat ~/apt_update_vs_upgrade.md

The command 'apt update' refreshes the list of available packages from the repositories.
It does not install or upgrade any software; it only updates the local package index.

The command 'apt upgrade' installs the latest versions of packages that are already installed on the system.
In short, 'apt update' prepares your system for upgrades, and 'apt upgrade' actually performs them.
If you skip 'apt update', your upgrade may miss the newest versions.
```

Task 4: Install Visual Studio Code via snap

Explanation:

Install VS Code using the snap package manager and verify installation from the command line.

Steps with Required Screenshots:

1. Install VS Code via snap.

Screenshot: task4_snap_install.png

```
tooba65@ubuntu:~$ sudo snap install --classic code

[sudo] password for tooba65:
Sorry, try again.
[sudo] password for tooba65:
error: cannot perform the following tasks:
- Download snap "code" (211) from channel "stable" (Get "https://api.snapcraft.io on 127.0.0.53:53: server misbehaving)
tooba65@ubuntu:~$ sudo snap install --classic code
code 7d842fb8 from Visual Studio Code (vscode) installed
```

2. Verify installation using snap list code.

Screenshot: task4_snap_list.png

```
tooba65@ubuntu:~$ snap list code
```

Name	Version	Rev	Tracking	Publisher	Notes
code	7d842fb8	211	latest/stable	vscode 	classic

3. Check version or info.

Screenshot: task4_code_version_or_info.png

```
tooba65@ubuntu:~$ code --version
1.105.1
7d842fb85a0275a4a8e4d7e040d2625abbf7f084
x64
```

4. Show binary location in /snap/bin.

Screenshot: task4_snap_bin_location.png

```
tooba65@ubuntu:~$ ls -l /snap/bin | grep code
lrwxrwxrwx 1 root root 13 Oct 24 05:53 code -> /usr/bin/snap
lrwxrwxrwx 1 root root 13 Oct 24 05:53 code.url-handler -> /usr/bin/snap
```

Task 5: Install XFCE GUI + XRDP and launch VS Code

Explanation:

Install a lightweight GUI and remote desktop access service (XRDP), then verify by launching VS Code in the GUI.

Steps with Required Screenshots:

1. Update and upgrade packages.

Screenshot: task5_update.png

```

  ⌕ Home ✖  🐧 Ubuntu 64-bit ✖
Ign:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:2 http://archive.ubuntu.com/ubuntu noble InRelease
Ign:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:2 http://archive.ubuntu.com/ubuntu noble InRelease
Ign:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.4 kB]
Fetched 268 kB in 28s (9,411 B/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates
  default-jre-headless fontconfig fonts-dejavu-extra gsettings-desktop-schemas gtk-update-icon-ca
  libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni
  libavahi-common-data libavahi-common3 libcairo-gobject2 libcairo2 libcolord2 libcups2t64 libdat
  libgdk-pixbuf2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgl1 libgl1-mesa-dri
  libgtk-3-bin libgtk-3-common libharfbuzz0b libice6 liblcms2-2 libllvm20 libpango-1.0-0 libpango
  libpaxman-1-0 librsvg2-2 librsvg2-common libsm6 libthai-data libthai0 libvulkan1 libwayland-cli
  libx11-xcb1 libxaw7 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-randr0 libxcb-render0 libx
  libxcomposite1 libxcursor1 libxdamage1 libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1 libxm
  libxv1 libxxf86dga1 libxxf86vm1 mesa-libgallium mesa-vulkan-drivers openjdk-21-jre openjdk-21-j
  x11-utils
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.

```

2. Install XFCE desktop environment.

Screenshot: task5_xfce_install.png

```
aspell-autobuildhash: processing: en [en_GB-ise-wo_accents-only].
aspell-autobuildhash: processing: en [en_GB-ize-w_accents-only].
aspell-autobuildhash: processing: en [en_GB-ize-wo_accents-only].
aspell-autobuildhash: processing: en [en_GB-variant_0].
aspell-autobuildhash: processing: en [en_GB-variant_1].
aspell-autobuildhash: processing: en [en_US-w_accents-only].
aspell-autobuildhash: processing: en [en_US-wo_accents-only].
Processing triggers for sgml-base (1.31) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-85-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-86-gene

Restarting the system to load the new kernel will not be handled automatically, so you

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
tooba65 @ session #1: login[920]
tooba65 @ session #3: sshd[1276]
tooba65 @ user manager service: systemd[1204]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

3. Install and enable XRDP.

Screenshot: task5_xrdp_enable.png

```

Created symlink /etc/systemd/system/multi-user.target.wants/xrdp-sesman.service → /usr/lib/systemd/system/
Created symlink /etc/systemd/system/multi-user.target.wants/xrdp.service → /usr/lib/systemd/system/xrdp.se
Setting up pipewire-module-xrdp (0.2-2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
6.8.0-85-generic
Diagnostics:
The currently running kernel version is not the expected kernel version 6.8.0-86-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider reb
Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
tooba65 @ session #1: login[920]
tooba65 @ session #3: sshd[1276]
tooba65 @ user manager service: systemd[1204]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
tooba65@ubuntu:~$ sudo systemctl enable --now xrdp
Synchronizing state of xrdp.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable xrdp

```

4. Check XRDP status.

Screenshot: task5_xrdp_status.png

```

tooba65@ubuntu:~$ sudo systemctl status xrdp
● xrdp.service - xrdp daemon
   Loaded: loaded (/usr/lib/systemd/system/xrdp.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-10-24 06:47:44 UTC; 3min 1s ago
     Docs: man:xrdp(8)
           man:xrdp.ini(5)
   Main PID: 50646 (xrdp)
      Tasks: 1 (limit: 4548)
    Memory: 868.0K (peak: 1.5M)
         CPU: 34ms
    CGroup: /system.slice/xrdp.service
            └─50646 /usr/sbin/xrdp

Oct 24 06:47:43 ubuntu systemd[1]: Starting xrdp.service - xrdp daemon...
Oct 24 06:47:43 ubuntu xrdp[50644]: [INFO ] address [0.0.0.0] port [3389] mode 1
Oct 24 06:47:43 ubuntu xrdp[50644]: [INFO ] listening to port 3389 on 0.0.0.0
Oct 24 06:47:43 ubuntu xrdp[50644]: [INFO ] xrdp_listen_pp done
Oct 24 06:47:43 ubuntu systemd[1]: xrdp.service: Can't open PID file /run/xrdp/xrdp.pid (yet?) after start: No such file or directory
Oct 24 06:47:44 ubuntu systemd[1]: Started xrdp.service - xrdp daemon.
Oct 24 06:47:45 ubuntu xrdp[50646]: [INFO ] starting xrdp with pid 50646
Oct 24 06:47:45 ubuntu xrdp[50646]: [INFO ] address [0.0.0.0] port [3389] mode 1
Oct 24 06:47:45 ubuntu xrdp[50646]: [INFO ] listening to port 3389 on 0.0.0.0
Oct 24 06:47:45 ubuntu xrdp[50646]: [INFO ] xrdp_listen_pp done

```

5. Configure XFCE session.

Screenshot: task5_xsession.png

```

tooba65@ubuntu:~$ echo xfce4-session > ~/.xsession
tooba65@ubuntu:~$

```

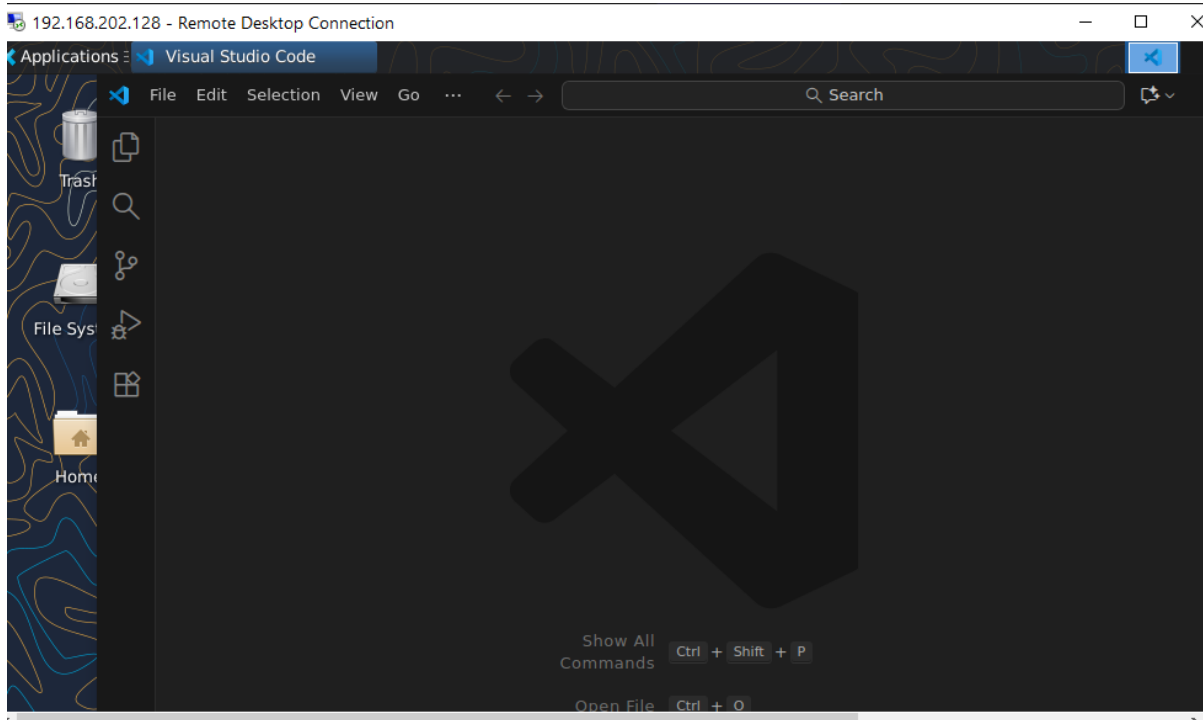
6. Connect via Remote Desktop.

Screenshot: task5_rdp_connect.png



7. Launch VS Code in GUI.

Screenshot: task5_vscode_launch.png



Task 6: Install lightdm-gtk-greeter and GUI verification

Explanation:

Fix GUI login issues, configure LightDM, control boot behavior, and test starting/stopping GUI manually.

Steps with Required Screenshots:

1. Install LightDM and greeter.

Screenshot: task6_lightdm_install.png

```
Setting up gnome-themes-extra-data (3.28-2ubuntu5) ...
Setting up gnome-accessibility-themes (3.28-2ubuntu5) ...
Setting up gtk2-engines-pixbuf:amd64 (2.24.33-4ubuntu1.1) ...
Setting up lightdm-gtk-greeter (2.0.9-0ubuntu3) ...
update-alternatives: using /usr/share/xgreeters/lightdm-gtk-greeter.desktop to provide /usr/share/xgreeters/lightdm-greeter
Setting up gnome-themes-extra:amd64 (3.28-2ubuntu5) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  6.8.0-85-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 6.8.0-86-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
tooba65 @ session #1: login[920]
tooba65 @ session #3: sshd[1276]
tooba65 @ user manager service: systemd[1204]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

2. Create configuration file.

Screenshot: task6_lightdm_config.png

```
tooba65@ubuntu:~$ sudo mkdir -p /etc/lightdm/lightdm.conf.d
tooba65@ubuntu:~$ echo -e "[Seat:*]\ngreeter-session=lightdm-gtk-greeter\nuser-session=xfce\nautologin-user=tim\nautologin-user-timeout=0" > /etc/lightdm/lightdm.conf.d/50-xfce.conf
[Seat:*]
greeter-session=lightdm-gtk-greeter
user-session=xfce
autologin-user=tim
autologin-user-timeout=0
```

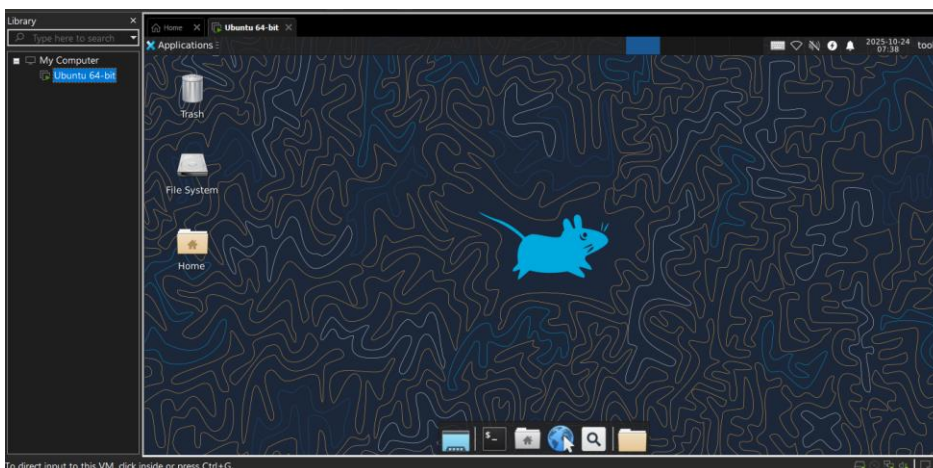
3. Clean session files and permissions.

Screenshot: task6_lightdm_cleanup.png

```
tooba65@ubuntu:~$ sudo rm -f /var/lib/lightdm/.Xauthority
tooba65@ubuntu:~$ sudo rm -f ~/.Xauthority
tooba65@ubuntu:~$ sudo rm -rf ~/.cache/sessions
tooba65@ubuntu:~$ sudo chown -R $USER:$USER /home/$USER
```

4. Restart LightDM.

Screenshot: task6_lightdm_restart.png



5. Enable GUI at boot and reboot.

Screenshots: task6_gui_enable_boot.png, task6_after_reboot_gui.png

```
tooba65@ubuntu:~$ sudo systemctl restart lightdm

tooba65@ubuntu:~$ sudo systemctl enable lightdm
Synchronizing state of lightdm.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable lightdm
The unit files have no installation config (WantedBy=, RequiredBy=, UpheldBy=,
Also=, or Alias= settings in the [Install] section, and DefaultInstance= for
template units). This means they are not meant to be enabled or disabled using systemctl.

Possible reasons for having these kinds of units are:
• A unit may be statically enabled by being symlinked from another unit's
  .wants/, .requires/, or .upholds/ directory.
• A unit's purpose may be to act as a helper for some other unit which has
  a requirement dependency on it.
• A unit may be started when needed via activation (socket, path, timer,
  D-Bus, udev, scripted systemctl call, ...).
• In case of template units, the unit is meant to be enabled with some
  instance name specified.
```

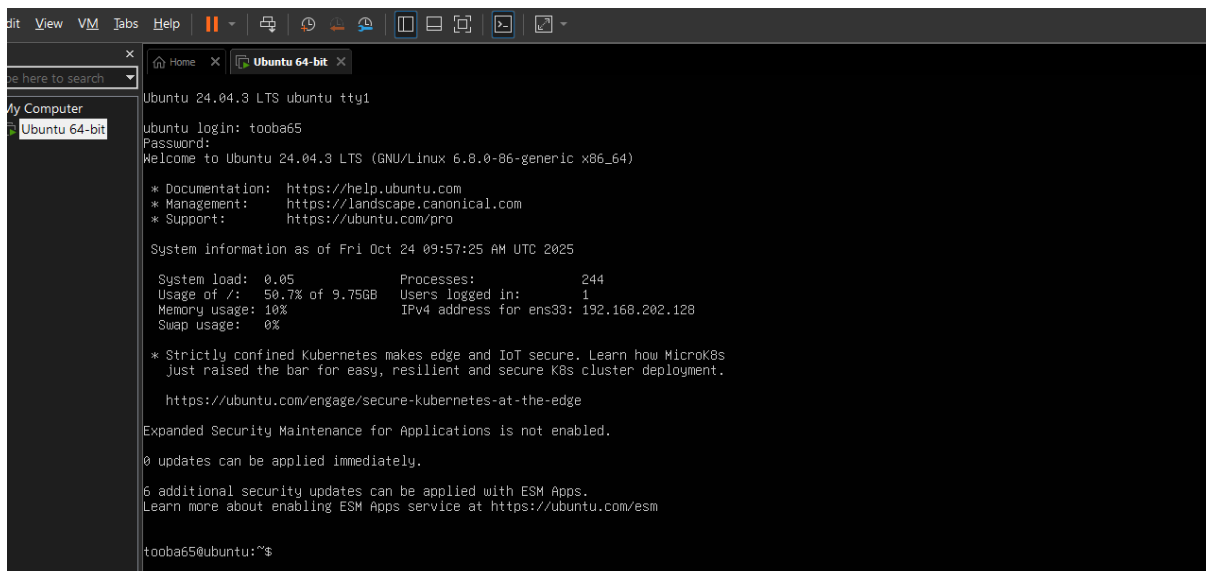
```
tooba65@ubuntu:~$ sudo systemctl set-default graphical.target
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/graphical.target.
```

6. Disable GUI at boot and reboot.

Screenshots: task6_gui_disable_boot.png, task6_after_reboot_cli.png

```
tooba65@ubuntu:~$ sudo systemctl set-default multi-user.target
[sudo] password for tooba65:
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target.
```

```
tooba65@ubuntu:~$ sudo systemctl disable lightdm
Synchronizing state of lightdm.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable lightdm
Removed "/etc/systemd/system/display-manager.service".
```



The screenshot shows a terminal window with the following content:

```
Ubuntu 24.04.3 LTS ubuntu tty1
ubuntu login: tooba65
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Fri Oct 24 09:57:25 AM UTC 2025

System load:  0.05          Processes:      244
Usage of /:   50.7% of 9.75GB Users logged in:  1
Memory usage: 10%          IPv4 address for ens33: 192.168.202.128
Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.
   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

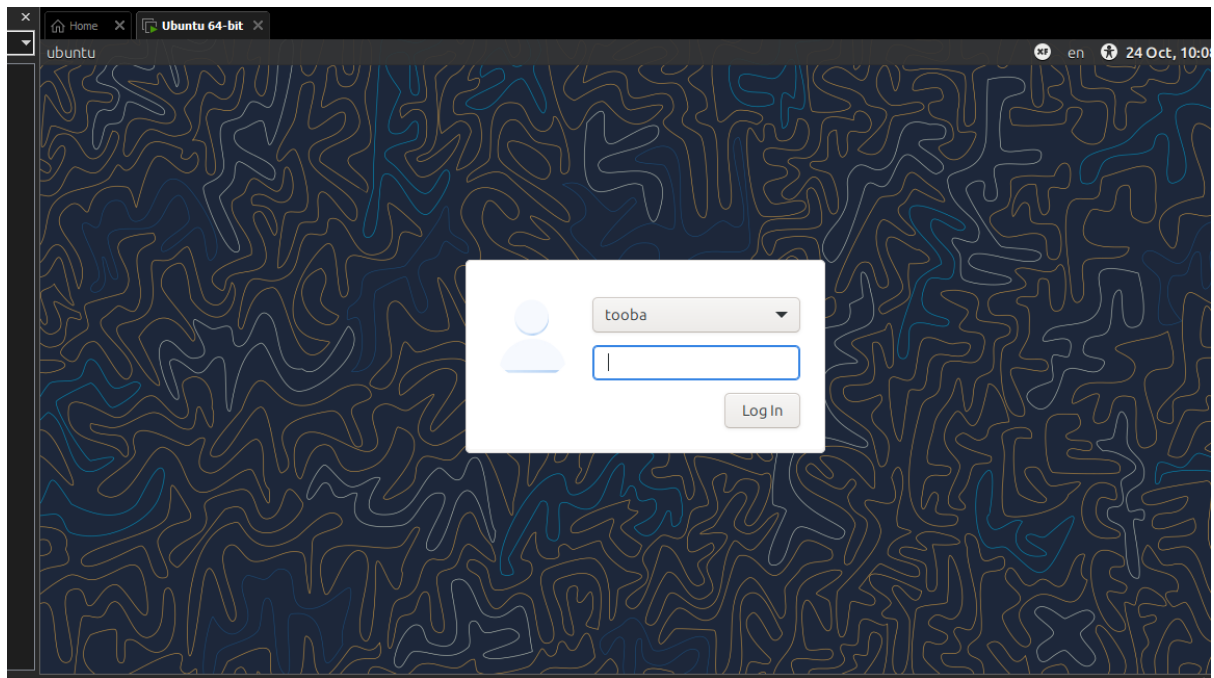
6 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

tooba65@ubuntu:~$
```

7. Start and stop GUI manually.

Screenshots: task6_gui_start.png, task6_gui_stop.png

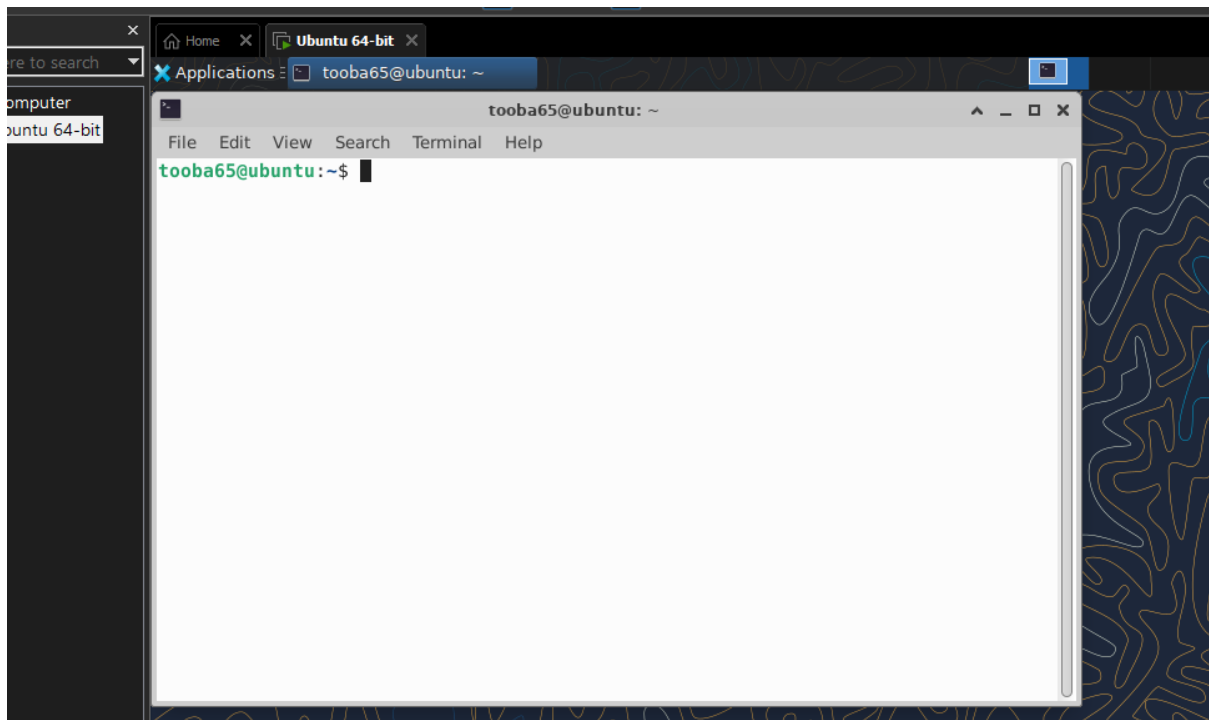
```
tooba65@ubuntu:~$ sudo systemctl start lightdm
[sudo] password for tooba65:
```



```
tooba65@ubuntu:~$ sudo systemctl stop lightdm
```

8. Start GUI and launch VS Code.

Screenshots: task6_gui_start_command.png, task6_vscode_launch.png



Task 7: Install Google Chrome using apt source and key

Explanation:

Add Google's apt repository and signing key to install Chrome. Demonstrate both error and successful installation.

Steps with Required Screenshots:

1. Attempt installation (expected error).

Screenshot: task7_install_chrome_error.png

```
tooba65@ubuntu:~$ sudo apt install google-chrome-stable -y
[sudo] password for tooba65:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package google-chrome-stable
```

2. Inspect apt directories and lists.

Screenshots: task7_ls_etc_apt.png, task7_cat_sources_list.png, task7_ls_sources_list_d.png, task7_cat_ubuntu_sources.png

```
tooba65@ubuntu:~$ ls -la /etc/apt
total 48
drwxr-xr-x  9 root root  4096 Oct  2 08:26 .
drwxr-xr-x 142 root root 12288 Oct 24 07:02 ..
drwxr-xr-x  2 root root  4096 Oct 24 05:14 apt.conf.d
drwxr-xr-x  2 root root  4096 Mar 31 2024 auth.conf.d
drwxr-xr-x  2 root root  4096 Mar 31 2024 keyrings
drwxr-xr-x  2 root root  4096 Aug  5 17:14 preferences.d
drwxr-xr-x  2 root root  4096 Aug  5 17:14 preferences.d.save
-rw-r--r--  1 root root    70 Oct  2 04:00 sources.list
drwxr-xr-x  2 root root  4096 Oct  2 04:00 sources.list.d
drwxr-xr-x  2 root root  4096 Aug  5 17:01 trusted.gpg.d
```

```
tooba65@ubuntu:~$ cat /etc/apt/sources.list
# Ubuntu sources have moved to /etc/apt/sources.list.d/ubuntu.sources
```

```
tooba65@ubuntu:~$ ls -la /etc/apt/sources.list.d/
total 16
drwxr-xr-x 2 root root 4096 Oct  2 04:00 .
drwxr-xr-x 9 root root 4096 Oct  2 08:26 ..
-rw-r--r-- 1 root root  383 Oct  2 08:26 ubuntu.sources
-rw-r--r-- 1 root root 2552 Aug  5 17:02 ubuntu.sources.curtin.orig
```

```
tooba65@ubuntu:~$ cat /etc/apt/sources.list.d/ubuntu.sources
Types: deb
URIs: http://archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

3. Add Chrome repository and key.

Screenshots: task7_edit_ubuntu_sources.png, task7_add_key.png

```
tooba65@ubuntu:~$ cat /etc/apt/sources.list.d/ubuntu.sources
Types: deb
URIs: http://archive.ubuntu.com/ubuntu/
Suites: noble noble-updates noble-backports
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://security.ubuntu.com/ubuntu/
Suites: noble-security
Components: main restricted universe multiverse
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg

Types: deb
URIs: http://dl.google.com/linux/chrome/deb/
Suites: stable
Components: main
Architectures: amd64
Signed-By: /etc/apt/keyrings/google.gpg
```

```
tooba65@ubuntu:~$ sudo mkdir -p /etc/apt/keyrings
[sudo] password for tooba65:
```

```
tooba65@ubuntu:~$ curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg
```

4. Update apt and install Chrome.

Screenshots: task7_apr_update.png, task7_install_chrome.png

```
tooba65@ubuntu:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://dl.google.com/linux/chrome/deb stable InRelease [1,825 B]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:5 http://dl.google.com/linux/chrome/deb stable/main amd64 Packages [1,214 B]
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 129 kB in 2s (79.4 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

task7_install_chrome.png:

```
Unpacking google-chrome-stable (141.0.7390.122-1) ...
Selecting previously unselected package fonts-liberation-sans-narrow.
Preparing to unpack .../fonts-liberation-sans-narrow_1%3a1.07.6-4_all.deb ...
Unpacking fonts-liberation-sans-narrow (1:1.07.6-4) ...
Setting up fonts-liberation (1:2.1.5-3) ...
Setting up fonts-liberation-sans-narrow (1:1.07.6-4) ...
Setting up google-chrome-stable (141.0.7390.122-1) ...
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/x-www-browser (x-www-browser)
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/gnome-www-browser (gnome-www-browser)
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/google-chrome (google-chrome)
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for fontconfig (2.15.0-1.1ubuntu2) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
tooba65@ubuntu:~$
```

```
no VM guests are running outdated hypervisor  
tooba65@ubuntu:~$ google-chrome --version  
Google Chrome 141.0.7390.122
```

5. Perform cleanup and use alternate preferred method.

Screenshots: task7_alternate_remove.png, task7_alternate_edit.png,
task7_remove_key.png

```
tooba65@ubuntu:~$ sudo apt remove google-chrome-stable -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following packages were automatically installed and are no longer required:  
  ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libatk-wr  
  libgif7 openjdk-21-jre openjdk-21-jre-headless  
Use 'sudo apt autoremove' to remove them.  
The following packages will be REMOVED:  
  google-chrome-stable  
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.  
After this operation, 396 MB disk space will be freed.  
(Reading database ... 180499 files and directories currently installed.)  
Removing google-chrome-stable (141.0.7390.122-1) ...  
Processing triggers for desktop-file-utils (0.27-2build1) ...  
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...  
Rebuilding /usr/share/applications/bamf-2.index...  
tooba65@ubuntu:~$
```

task7_alternate_edit.png:

```
tooba65@ubuntu:~$ cat /etc/apt/sources.list.d/ubuntu.sources  
Types: deb  
URIs: http://archive.ubuntu.com/ubuntu/  
Suites: noble noble-updates noble-backports  
Components: main restricted universe multiverse  
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg  
  
Types: deb  
URIs: http://security.ubuntu.com/ubuntu/  
Suites: noble-security  
Components: main restricted universe multiverse  
Signed-By: /usr/share/keyrings/ubuntu-archive-keyring.gpg
```

```
tooba65@ubuntu:~$ sudo rm -f /etc/apt/keyrings/google.gpg  
tooba65@ubuntu:~$
```

6. Add single-line Chrome list and key again.

Screenshots: task7_create_google_chrome_list.png,
task7_list_sources_after_create.png, task7_add_key_alt.png

```
tooba65@ubuntu:~$ echo "deb [arch=amd64 signed-by=/etc/apt/keyrings/google.gpg] http://dl.google.com/linux/chrome/deb/ s  
table main" | sudo tee /etc/apt/sources.list.d/google-chrome.list > /dev/null
```

task7_list_sources_after_create.png:

```
tooba65@ubuntu:~$ ls -la /etc/apt/sources.list.d/
total 20
drwxr-xr-x 2 root root 4096 Oct 24 14:13 .
drwxr-xr-x 9 root root 4096 Oct  2 08:26 ..
-rw-r--r-- 1 root root  107 Oct 24 14:20 google-chrome.list
-rw-r--r-- 1 root root  386 Oct 24 14:13 ubuntu.sources
-rw-r--r-- 1 root root 2552 Aug  5 17:02 ubuntu.sources.curtin.orig
```

task7_add_key_alt.png:

```
tooba65@ubuntu:~$ sudo mkdir -p /etc/apt/keyrings
tooba65@ubuntu:~$ curl -fsSL https://dl.google.com/linux/linux_signing_key.pub | sudo gpg --dearmor -o /etc/apt/keyrings/google.gpg
```

7. Update and install again.

Screenshots: task7_apt_update_alt.png, task7_install_chrome_alt.png

```
tooba65@ubuntu:~$ sudo apt update
Hit:1 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

task7_install_chrome_alt.png:

```
tooba65@ubuntu: ~
Preparing to unpack .../google-chrome-stable_141.0.7390.122-1_amd64.deb ...
Unpacking google-chrome-stable (141.0.7390.122-1) ...
Setting up google-chrome-stable (141.0.7390.122-1) ...
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/x-www-browser (x-www-browser)
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/gnome-www-browser (gnome-www-browser)
update-alternatives: using /usr/bin/google-chrome-stable to provide /usr/bin/google-chrome (google-chrome)
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.
After this operation, 396 MB of additional disk space will be used.
No services need to be restarted.chrome/deb stable/main amd64 google-chrome-stable amd64 141.0.7390.122-1
Fetched 120 MB in 1min 28s (1,364 kB/s)
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
tooba65@ubuntu:~$ google-chrome --version
Google Chrome 141.0.7390.122
```

Task 8: Install applications via PPA (Audacity & OBS)

Explanation:

Learn how to add Personal Package Archives (PPAs) and install GUI applications.

Steps with Required Screenshots:

1. Add Audacity PPA and update apt.

Screenshots: task8_add_ppa_audacity.png, task8_apt_update_audacity.png

```
tooba65@ubuntu:~$ sudo add-apt-repository ppa:ubuntuhandbook1/audacity -y
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu/
Suites: noble
Components: main
'
Description:
Unofficial build of Audacity audio editor

For help, please use Audacity forum: http://forum.audacityteam.org/

If the packages here are helpful, you may buy me a coffee:

https://ko-fi.com/ubuntuhandbook1
More info: https://launchpad.net/~ubuntuhandbook1/+archive/ubuntu/audacity
Adding repository.
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:6 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease [18.1 kB]
Get:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble/main amd64 Packages [1,064 B]
Get:8 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble/main Translation-en [492 B]
Fetched 19.6 kB in 2s (10.6 kB/s)
Reading package lists... Done
```

task8_apt_update_audacity.png:

```
Reading package lists... Done
tooba65@ubuntu:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:6 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

2. Install Audacity and verify/launch.

Screenshots: task8_install_audacity.png, task8_audacity_launch.png or task8_audacity_version.png

```
tooba65@ubuntu: ~
Setting up libqt5qmlmodels5:amd64 (5.15.13+dfsg-1ubuntu0.1) ...
Setting up libqt5widgets5:amd64 (5.15.13+dfsg-1ubuntu1) ...
Setting up libqt5svg5:amd64 (5.15.13-1) ...
Setting up audacity (3.7.5-0build1~ubuntu24.04) ...
Setting up qt5-gtk-platformtheme:amd64 (5.15.13+dfsg-1ubuntu1) ...
Setting up libqt5waylandclient5:amd64 (5.15.13-1) ...
Setting up libqt5quick5:amd64 (5.15.13+dfsg-1ubuntu0.1) ...
Setting up libqt5waylandcompositor5:amd64 (5.15.13-1) ...
Setting up qtwayland5:amd64 (5.15.13-1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for shared-mime-info (2.4-4) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
tooba65@ubuntu:~$
```

task8_audacity_version.png:

```
tooba65@ubuntu:~$ audacity --version
tooba65@ubuntu:~$ dpkg -l | grep audacity
ii  audacity                    3.7.5-0build1~ubuntu24.04      amd64
ii  audacity-data               3.7.5-0build1~ubuntu24.04      all
tooba65@ubuntu:~$ which audacity
/usr/bin/audacity
tooba65@ubuntu:~$ apt show audacity
Package: audacity
Version: 3.7.5-0build1~ubuntu24.04
Priority: optional
Section: sound
Maintainer: Jhonny Oliveira <jhonny.oliveira@gmail.com>
Original-Maintainer: Debian Multimedia Maintainers <debian-multimedia@lists.debian.org>
```

3. Add OBS Studio PPA and update apt.

Screenshots: task8_add_ppa_obs.png, task8_apt_update_obs.png

```
tooba65@ubuntu:~$ sudo add-apt-repository ppa:obsproject/obs-studio -y
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu/
Suites: noble
Components: main
'
Description:
Latest stable release of OBS Studio
More info: https://launchpad.net/~obsproject/+archive/ubuntu/obs-studio
Adding repository.
Hit:1 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:6 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease [17.8 kB]
Hit:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Get:8 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main amd64 Packages [1,172 B]
Get:9 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble/main Translation-en [160 B]
Fetched 19.1 kB in 2s (9,261 B/s)
Reading package lists... Done
```

task8_apt_update_obs.png:

```
tooba65@ubuntu:~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:6 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease
Hit:7 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

4. Install and verify/launch OBS.

Screenshots: task8_install_obs.png, task8_obs_launch.png or task8_obs_version.png

```

tooba65@ubuntu: ~
Setting up libmbd1:amd64 (2.28.8-1) ...
Setting up libqt6wlshellintegration6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandcompositor6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandeglcompositorhwintegration6:amd64 (6.4.2-5build3) ...
Setting up libqt6waylandeglclienthwintegration6:amd64 (6.4.2-5build3) ...
Setting up libpocketsphinx3:amd64 (0.8.0+real5prealpha+1-15ubuntu5) ...
Setting up qt6-wayland:amd64 (6.4.2-5build3) ...
Setting up libavfilter9:amd64 (7:6.1.1-3ubuntu5) ...
Setting up libavdevice60:amd64 (7:6.1.1-3ubuntu5) ...
Setting up obs-studio (32.0.0-0obsproject1~noble) ...
Processing triggers for bamfdaemon (0.5.6+22.04.20220217-0ubuntu5) ...
Rebuilding /usr/share/applications/bamf-2.index...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.0%

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```

task8_obs_version.png:

```

tooba65@ubuntu:~$ obs --version
OBS Studio - 32.0.0
tooba65@ubuntu:~$

```

Task 9: Create a Kubernetes YAML using Vim

Explanation:

Use Vim to create a Kubernetes Pod manifest and save it in your Lab5 folder.

Steps with Required Screenshots:

1. Check if Vim is installed.
Screenshot: task9_vim_check.png

4. Save and verify file creation.

Screenshot: task9_k8s_saved.png

```
tooba65@ubuntu:~/Lab5$ ls -la
total 12
drwxrwxr-x  2 tooba65 tooba65 4096 Oct 24 15:34 .
drwxr-x--- 24 tooba65 tooba65 4096 Oct 24 15:34 ..
-rw-rw-r--  1 tooba65 tooba65  285 Oct 24 15:34 k8s-sample.yaml
tooba65@ubuntu:~/Lab5$
```

Task 10: Edit the YAML – add annotation and discard change

Explanation:

Practice editing files, saving permanent changes, and discarding temporary edits in Vim.

Steps with Required Screenshots:

1. Add annotation under metadata and save.

Screenshot: task10_verify_annotation.png

```
tooba65@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
        restartPolicy: Always
tooba65@ubuntu:~/Lab5$
```

2. Reopen, add temporary comment, and exit without saving.

Screenshot: task10_verify_entering_temp_data.png

```
tooba65@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
        restartPolicy: Always
        # temp: do-not-keep
~
~
~
~
```

3. Verify comment not saved.

Screenshot: task10_verify_no_temp_comment.png

```
tooba65@ubuntu:~/Lab5$ vim k8s-sample.yaml
tooba65@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
        restartPolicy: Always
```

Task 11: Vim editing practice – delete, undo, navigation

Explanation:

Practice delete (dd), undo (u), numeric delete, and navigation commands.

Steps with Required Screenshots:

1. Delete a line and undo.

Screenshot: task11 dd delete and undo.png

```
tooba65@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
          restartPolicy: Always
```

I undo the line after pressing dd immediately:

```
tooba65@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
          restartPolicy: Always
~
~
~
~
~
~
~
~
~
~
1 more line; before #1 29 seconds ago
```

```
tooba65@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
          restartPolicy: Always
```

2. Delete multiple lines and undo.
Screenshot: task11_delete3_and_undo.png

```

tooba65@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      restartPolicy: Always

```

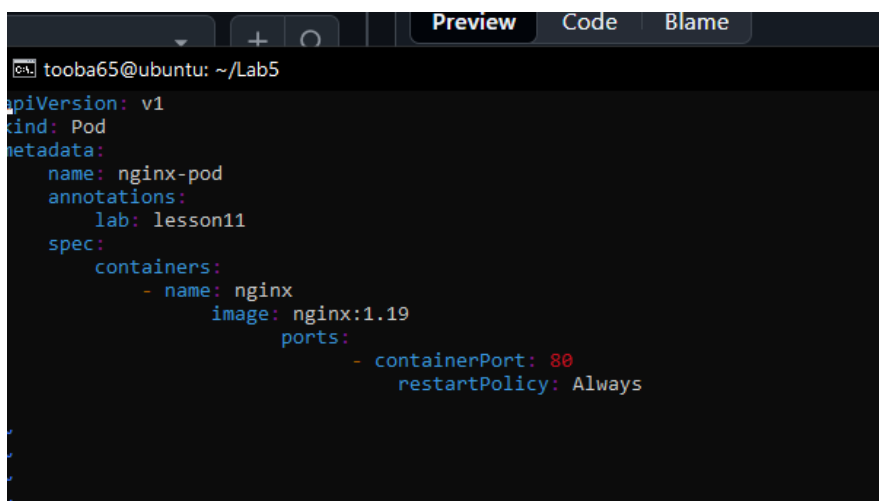
```

tooba65@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
      restartPolicy: Always

```

- Practice navigation (1G, G, \$, 0).

Screenshots: task11_line1.png



```

tooba65@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
      restartPolicy: Always

```

task11_navigation.png:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always
```

```
tooba65@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always
```

```
tooba65@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
  spec:
    containers:
      - name: nginx
        image: nginx:1.19
        ports:
          - containerPort: 80
            restartPolicy: Always
```

Task 12: Vim search, substitution, and undo

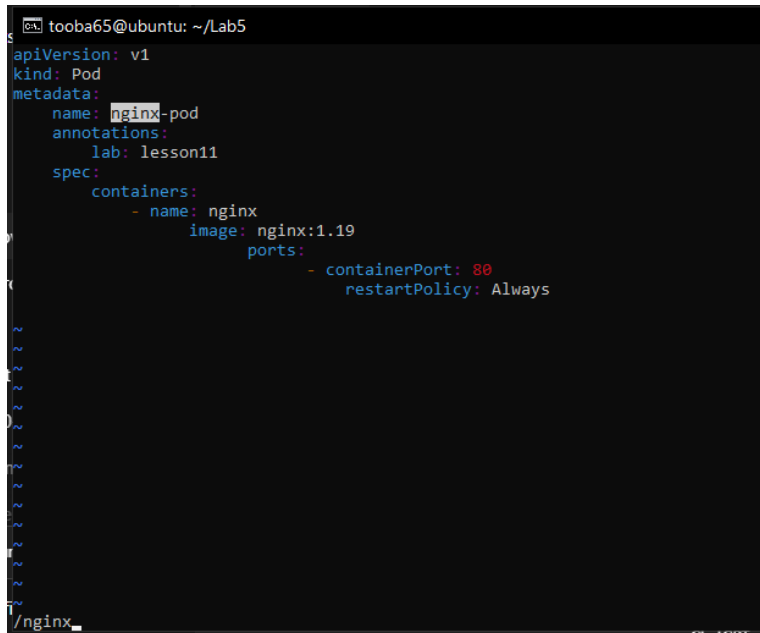
Explanation:

Learn searching, navigating matches, substituting text, and undoing changes in Vim.

Steps with Required Screenshots:

1. Search for the word “nginx”.

Screenshot: task12_search_nginx.png

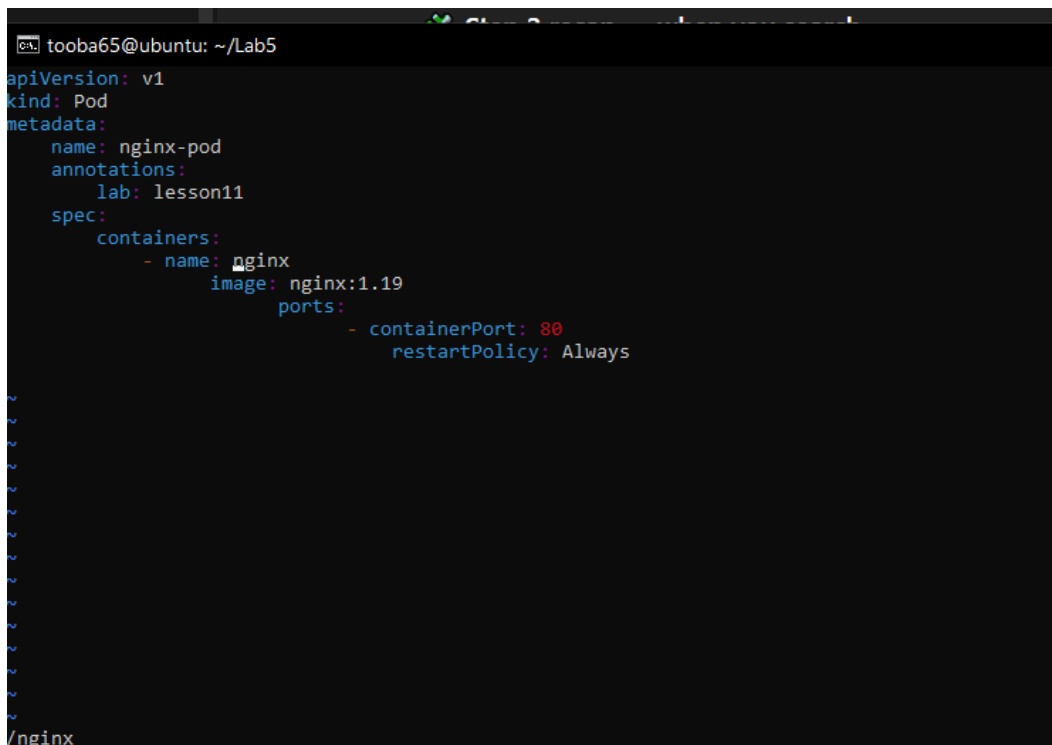


The screenshot shows a terminal window with the command prompt 'tooba65@ubuntu: ~/Lab5'. The user has opened a file in Vim, and the search command '/nginx' has been entered, highlighting the 'nginx' entry in the 'containers' list of a YAML file. The terminal output shows the following content:

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
          restartPolicy: Always
```

2. Move between matches with n and N.

Screenshot: task12_n_and_N_navigation.png



The screenshot shows the same terminal window as the previous one, but now the cursor is positioned on the 'nginx' entry in the 'containers' list. The user has used the 'n' key to move to the next match, which is also 'nginx'. The terminal output is the same as the previous screenshot.

```

tooba65@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
          restartPolicy: Always

```

3. Add extra “nginx” words and save.

Screenshot: task12_added_occurrences.png

```

tooba65@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  # nginx
  # nginx
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
          restartPolicy: Always

```

4. Cycle through matches with n.

Screenshot: task12_cycle_matches.png

```

tooba65@ubuntu: ~/Lab5
version: v1
kind: Pod
metadata:
  nginx
  nginx
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
          restartPolicy: Always

```

```

tooba65@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  # nginx
  # nginx
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
          restartPolicy: Always

```

```

tooba65@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  # nginx
  # nginx
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
          restartPolicy: Always

```

5. Replace all “nginx” with “webapp”.
- Screenshot:** task12_substitute_result.png

```

tooba65@ubuntu: ~/Lab5
apiVersion: v1
kind: Pod
metadata:
  # webapp
  # webapp
  name: webapp-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: webapp
      image: webapp:1.19
      ports:
        - containerPort: 80
          restartPolicy: Always

```

5 substitutions on 5 lines

6. Undo substitution and quit.

Screenshot: task12_undo_and_quit.png

```
tooba65@ubuntu:~/Lab5$ cat k8s-sample.yaml
apiVersion: v1
kind: Pod
metadata:
  # nginx
  # nginx
  name: nginx-pod
  annotations:
    lab: lesson11
spec:
  containers:
    - name: nginx
      image: nginx:1.19
      ports:
        - containerPort: 80
      restartPolicy: Always
```

Exam Evaluation Question – Docker Desktop Installation

Steps Performed

1. Updated the system packages to make sure all repositories and dependencies were current:
sudo apt update

```
tooba65@ubuntu: ~
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.4 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,226 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [211 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9,008 B]
Get:19 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,080 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [505 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [470 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:23 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [906 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,498 kB]
Get:26 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [203 kB]
Get:27 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.2 kB]
Get:28 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.4 kB]
Get:29 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [303 kB]
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.4 kB]
Get:33 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:34 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,140 B]
Get:35 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:36 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [11.0 kB]
Get:37 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 12.7 MB in 8s (1,535 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
13 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

sudo apt upgrade -y

```
tooba65@ubuntu:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-
```

2. Installed required packages for Docker and GUI support:

sudo apt install ca-certificates curl gnupg lsb-release -y

```
tooba65@ubuntu:~$ sudo apt install ca-certificates curl gnupg lsb-release -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.6).
curl set to manually installed.
gnupg is already the newest version (2.4.4-2ubuntu17.3).
gnupg set to manually installed.
lsb-release is already the newest version (12.0-2).
lsb-release set to manually installed.
The following packages were automatically installed and are no longer required:
  ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libatk-wrap
  libgif7 linux-headers-6.8.0-85 linux-headers-6.8.0-85-generic linux-image-6.8.0-85-g
  linux-modules-6.8.0-85-generic linux-modules-extra-6.8.0-85-generic linux-tools-6.8.
  linux-tools-6.8.0-85-generic openjdk-21-jre openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

3. Added Docker's official GPG key and repository:

```
sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o
/etc/apt/keyrings/docker.gpg
```

```
tooba65@ubuntu:~$ sudo mkdir -p /etc/apt/keyrings
tooba65@ubuntu:~$ curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
```

Then add the Docker repository:

```
echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \
https://download.docker.com/linux/ubuntu \
$(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

```
tooba65@ubuntu:~$ echo \
> "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \
> https://download.docker.com/linux/ubuntu \
> $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

4. Updated apt and installed Docker packages (engine, CLI, and Desktop):

```
sudo apt update
```

```
tooba65@ubuntu:~$ sudo apt update
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Hit:2 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [33.5 kB]
Hit:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:8 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease
Hit:9 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Fetched 82.0 kB in 2s (42.1 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

```
sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin dockercompose-plugin -y
```

```

tooba65@ubuntu:~$ sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni
  libgif7 linux-headers-6.8.0-85 linux-headers-6.8.0-85-generic linux-image-6.8.0-85-generic
  linux-modules-6.8.0-85-generic linux-modules-extra-6.8.0-85-generic linux-tools-6.8.0-85
  linux-tools-6.8.0-85-generic openjdk-21-jre openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  docker-ce-rootless-extras libslirp0 pigz slirp4netns
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin libslirp0
  pigz slirp4netns
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 105 MB of archives.
After this operation, 437 MB of additional disk space will be used.
Get:1 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 1.7.28-1~ubuntu.24.04~noble [31.9
MB]
Get:2 http://archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
tooba65@ubuntu:~$ docker --version
Docker version 28.5.1, build e180ab8

```

5. Downloaded and installed Docker Desktop for Linux:

wget <https://desktop.docker.com/linux/main/amd64/docker-desktop-amd64.deb>

```

tooba65@ubuntu:~$ wget https://desktop.docker.com/linux/main/amd64/docker-desktop-amd64.deb
--2025-11-02 11:41:49-- https://desktop.docker.com/linux/main/amd64/docker-desktop-amd64.deb
Resolving desktop.docker.com (desktop.docker.com)... 3.165.102.60, 3.165.102.21, 3.165.102.9, ...
Connecting to desktop.docker.com (desktop.docker.com)|3.165.102.60|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 421490404 (402M) [application/octet-stream]
Saving to: 'docker-desktop-amd64.deb'

docker-desktop-amd64.deb      100%[=====>] 401.96M  1.71MB/s   in 3m 37s
2025-11-02 11:45:26 (1.85 MB/s) - 'docker-desktop-amd64.deb' saved [421490404/421490404]

```

sudo apt install ./docker-desktop-amd64.deb -y

```

tooba65@ubuntu:~$ sudo apt install ./docker-desktop-amd64.deb -y
[sudo] password for tooba65:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'docker-desktop' instead of './docker-desktop-amd64.deb'
The following packages were automatically installed and are no longer required:
  ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java
  libgif7 openjdk-21-jre openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  cpu-checker ipxe-qemu ipxe-qemu-256k-compat-efi-roms libboost-iostreams1.83.0 libboost-threads
  libcacard0 libdaxctl1 libfdt1 libiscsi7 libndctl6 libpmem1 libpmemobj1 librados2 librbd1 lib
  libspice-server1 libsubid4 liburing2 libusbredirparser1t64 libvirglrenderer1 msr-tools ovmf

```

6. Started Docker Desktop application:

systemctl --user start docker-desktop

```

tooba65@ubuntu:~$ systemctl --user start docker-desktop
tooba65@ubuntu:~$

```

7. Verified Docker Desktop installation:

```

tooba65@ubuntu:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Sun 2025-11-02 11:30:52 UTC; 30min ago
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 8643 (dockerd)
      Tasks: 9
     Memory: 21.1M (peak: 22.1M)
        CPU: 805ms
    CGroup: /system.slice/docker.service
            └─8643 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Nov 02 11:30:51 ubuntu dockerd[8643]: time="2025-11-02T11:30:51.684304487Z" level=info msg="detected 127.0.0.1 as host address, assuming systemd-resolved, so using resolv.conf: dns=127.0.0.1, dnsFallback=127.0.0.1, dnsSuffix=."
Nov 02 11:30:51 ubuntu dockerd[8643]: time="2025-11-02T11:30:51.793408440Z" level=info msg="Creating a containerd client" address=/run/containerd/containerd.sock timeout=5s
Nov 02 11:30:51 ubuntu dockerd[8643]: time="2025-11-02T11:30:51.793408440Z" level=info msg="Loading containers: start."
Nov 02 11:30:52 ubuntu dockerd[8643]: time="2025-11-02T11:30:52.303329609Z" level=info msg="Loading containers: start."

```

- o Opened Applications → Docker Desktop
- o Confirmed the Docker icon appeared in the system tray.
- o Verified that Docker Engine was running with:
 - docker version

```

tooba65@ubuntu:~$ docker version
Client: Docker Engine - Community
 Version: 28.5.1
 API version: 1.51
 Go version: go1.24.8
 Git commit: e180ab8
 Built: Wed Oct 8 12:17:26 2025
 OS/Arch: linux/amd64
 Context: default
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Get "http://%2Frun%2Fdocker.sock/v1.51/version": dial unix /var/run/docker.sock: connect: permission denied

```

docker info

```

tooba65@ubuntu:~$ docker info
Client: Docker Engine - Community
 Version: 28.5.1
 Context: default
 Debug Mode: false
 Plugins:
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
     Version: v1.9.11
     Path: /home/tooba65/.docker/cli-plugins/docker-ai
  buildx: Docker Buildx (Docker Inc.)
     Version: v0.29.1-desktop.1
     Path: /home/tooba65/.docker/cli-plugins/docker-buildx
  compose: Docker Compose (Docker Inc.)
     Version: v2.40.2-desktop.1
     Path: /home/tooba65/.docker/cli-plugins/docker-compose
  debug: Get a shell into any image or container (Docker Inc.)
     Version: 0.0.45
     Path: /home/tooba65/.docker/cli-plugins/docker-debug
  desktop: Docker Desktop commands (Docker Inc.)
     Version: v0.2.0
     Path: /home/tooba65/.docker/cli-plugins/docker-desktop
  extension: Manages Docker extensions (Docker Inc.)
     Version: v0.2.31
     Path: /home/tooba65/.docker/cli-plugins/docker-extension
  init: Creates Docker-related starter files for your project (Docker Inc.)
     Version: v1.4.0
     Path: /home/tooba65/.docker/cli-plugins/docker-init
  mcp: Docker MCP Plugin (Docker Inc.)
     Version: v0.24.0

```

8. Install GUI (XFCE) Environment

Run this in your Ubuntu VM terminal:

```
sudo apt update
```

```
tooba65@ubuntu:~$ sudo apt update
[sudo] password for tooba65:
Hit:1 https://download.docker.com/linux/ubuntu noble InRelease
Hit:2 http://dl.google.com/linux/chrome/deb stable InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:7 https://ppa.launchpadcontent.net/obsproject/obs-studio/ubuntu noble InRelease
Hit:8 https://ppa.launchpadcontent.net/ubuntuhandbook1/audacity/ubuntu noble InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

`sudo apt install xfce4 xfce4-goodies -y`

```
tooba65@ubuntu:~$ sudo apt install xfce4 xfce4-goodies -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
xfce4 is already the newest version (4.18).
xfce4-goodies is already the newest version (4.18.2build1).
The following packages were automatically installed and are no longer required:
  ca-certificates-java default-jre-headless fonts-dejavu-extra java-common libat
  libgif7 openjdk-21-jre openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

This installs the XFCE desktop and some useful components (lightweight, works well in VMware). It may take a few minutes.

9. Verify or (Re)Enable XRDP for Remote GUI Access

Now let's ensure you can actually open and *see* the XFCE desktop GUI through VMware (or RDP if needed).

Run this command next:

`sudo systemctl enable --now xrdp`

```
tooba65@ubuntu:~$ sudo systemctl enable --now xrdp
Synchronizing state of xrdp.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable xrdp
```

Then check XRDP status:

`sudo systemctl status xrdp`

```
tooba65@ubuntu:~$ sudo systemctl status xrdp
● xrdp.service - xrdp daemon
   Loaded: loaded (/usr/lib/systemd/system/xrdp.service; enabled; preset: enabled)
   Active: active (running) since Sun 2025-11-02 11:00:40 UTC; 1h 59min ago
     Docs: man:xrdp(8)
           man:xrdp.ini(5)
  Main PID: 1292 (xrdp)
    Tasks: 1 (limit: 4545)
   Memory: 1.4M (peak: 1.9M)
      CPU: 78ms
   CGroup: /system.slice/xrdp.service
           └─1292 /usr/sbin/xrdp
```

10. Switch to GUI mode

Run this command in your terminal:

```
sudo systemctl set-default graphical.target sudo
systemctl isolate graphical.target
```

```
tooba65@ubuntu:~$ sudo systemctl set-default graphical.target
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/graphical.target.
tooba65@ubuntu:~$ sudo systemctl isolate graphical.target
tooba65@ubuntu:~$
```

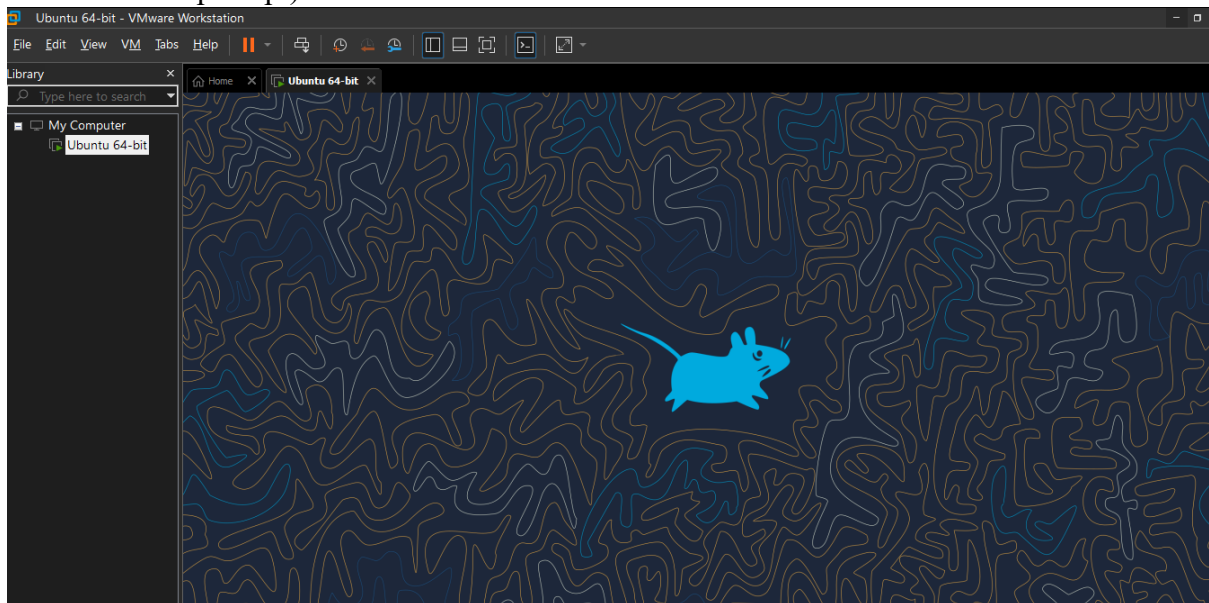
Then, wait a few seconds — your screen should change to a **login window** (the XFCE GUI login). If it doesn't, you can reboot manually:

```
sudo reboot
```

11. Started the GUI session manually when in CLI mode: `sudo systemctl start lightdm`

After this:

- In VMware, your VM will show a **login screen** (username/password prompt).

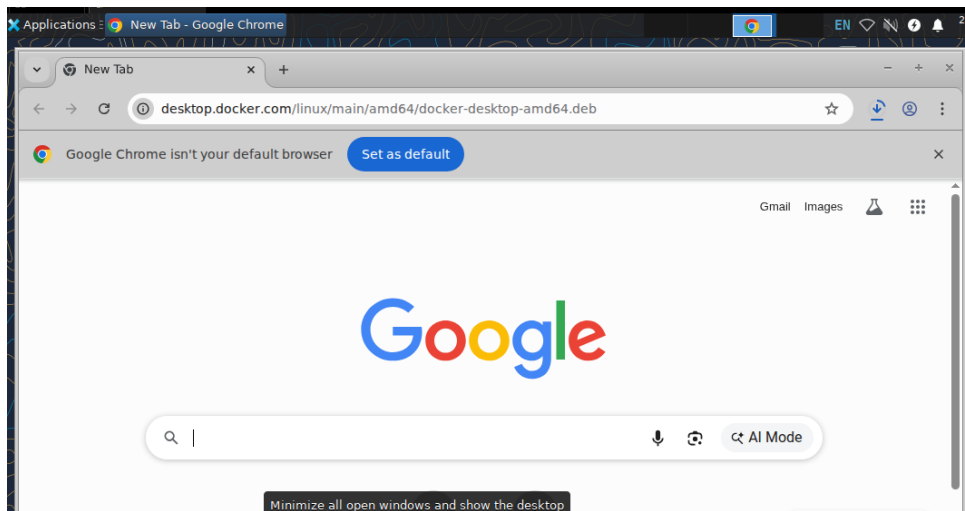


- Login as user **tooba**.
- When asked to choose a session, select **XFCE Session**.

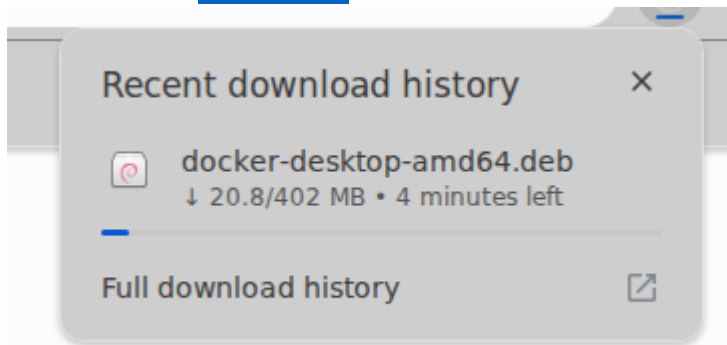
12. Download the Docker Desktop .deb package

Now that you're inside XFCE:

1. Click the **blue globe icon** (that's your web browser).



2. In the browser, go to given link and this automatically download the file:
<https://desktop.docker.com/linux/main/amd64/docker-desktop-amd64.deb>



13. Install Docker Desktop

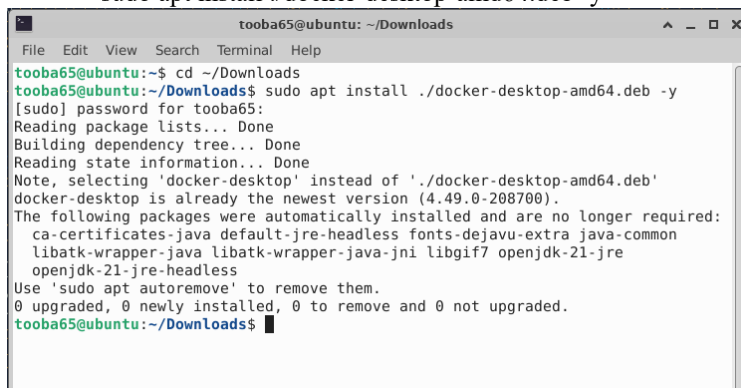
Run the following commands one by one **in your terminal** (you should still be in your home directory

~):

`cd ~/Downloads`

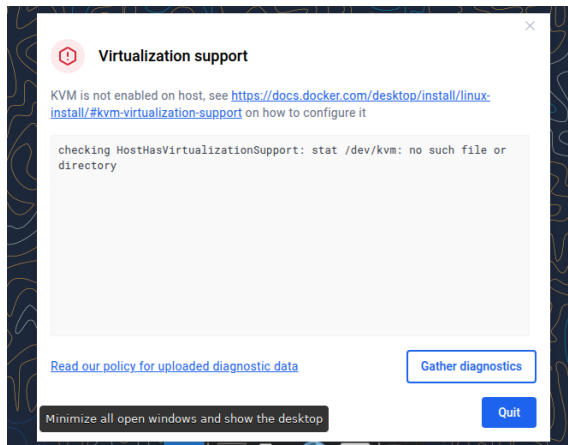
Then install it using:

`sudo apt install ./docker-desktop-amd64.deb -y`

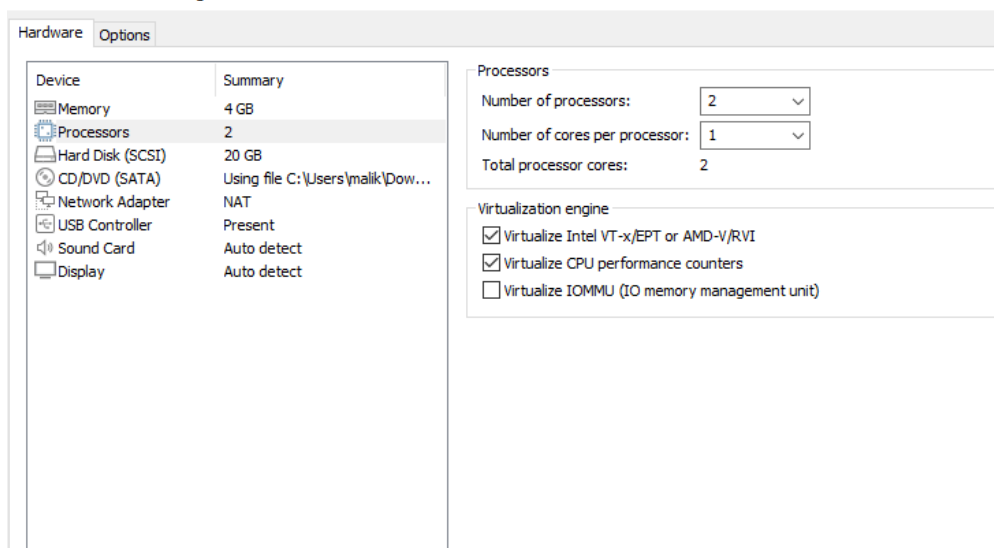


1. In your **Ubuntu GUI** (XFCE desktop), click the **Applications menu** (usually at the top-left or bottom-left corner).
2. Type **Docker Desktop** in the search bar.

3. Click on **Docker Desktop** to launch it.
Virtualization Error.



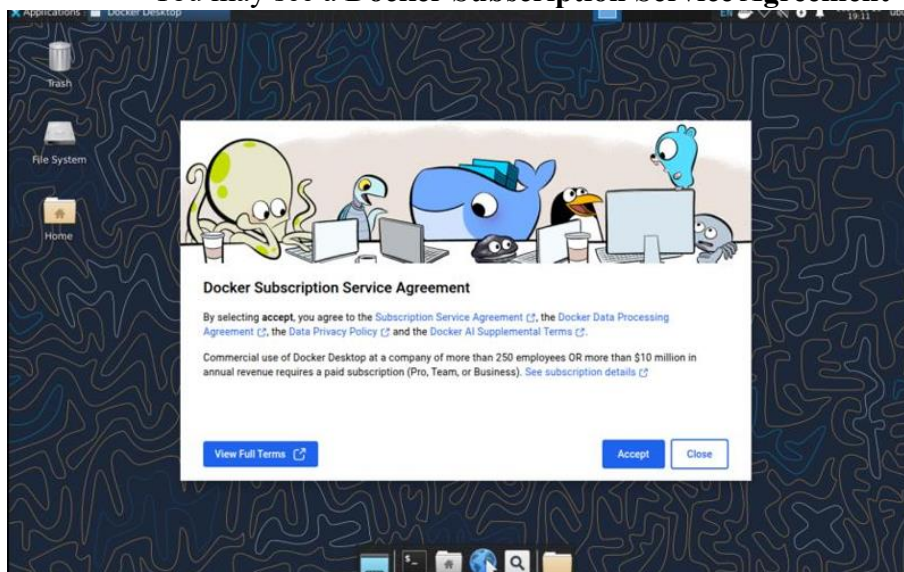
Virtual Machine Settings



Solution:

When it opens:

- You may see a **Docker Subscription Service Agreement** — click “Accept”.



- Then wait a bit while Docker Desktop starts up (it might take 30–60 seconds).
- Once you see the Docker whale icon or the main Docker Desktop window, that means it's running.

14. Captured a screenshot showing Docker Desktop running successfully.

